Case Report

Retroperitoneal Caecal Perforation Secondary to Abdominal Tuberculosis: A Rare Case Report

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Abstract

Background: Abdominal tuberculosis remains an important cause of acute and subacute abdominal emergencies in endemic regions. Ileocecal involvement is common, but caecal perforation, particularly with retroperitoneal extension, is rare and may be diagnostically challenging.

Case presentation: A 31-year-old woman, a known case of abdominal tuberculosis who had started anti-tubercular therapy 10 days earlier, presented with generalized abdominal pain, recurrent vomiting, obstipation and low-grade fever for six days. Clinical examination suggested peritonitis. Imaging showed pneumoperitoneum with bowel obstruction and computed tomography findings consistent with ileocecal tuberculosis. Colonoscopy revealed ulcerated mucosa with a tight narrowed segment, and biopsy was suggestive of tubercular colitis. Emergency exploratory laparotomy revealed a 1 x 1 cm caecal perforation, dense ileocecal adhesions and two completely non-passable strictures 30 cm and 70 cm proximal to the ileocecal junction. Right hemicolectomy with side-to-side jejuno-colic anastomosis and peritoneal toileting was performed. Histopathology showed granulomatous ileitis with epithelioid granulomas and Langhans-type giant cells, consistent with tuberculosis. Anti-tubercular therapy was continued post-operatively and the patient showed satisfactory clinical improvement.

Conclusion: Retroperitoneal caecal perforation is a rare but potentially life-threatening manifestation of abdominal tuberculosis. In endemic settings, tuberculosis should remain an important differential diagnosis in patients presenting with intestinal obstruction, perforation or atypical ileocecal pathology. Early surgical intervention, histopathological confirmation and continuation of appropriate anti-tubercular therapy are central to good outcome.

Keywords: Abdominal tuberculosis; Caecal perforation; Ileocecal tuberculosis; Intestinal obstruction; Right hemi colectomy; Anti-tubercular therapy

Introduction:

Abdominal tuberculosis may involve the gastrointestinal tract, peritoneum, mesenteric lymph nodes and solid abdominal viscera. Gastrointestinal disease most often affects the ileocecal region, where lymphoid tissue, relative stasis and absorptive activity favor involvement.1 Its clinical presentation is frequently nonspecific, with abdominal pain, fever, altered bowel habits, weight loss, obstruction or a palpable mass, and it may mimic Crohn disease, malignancy or other inflammatory conditions.

Although abdominal tuberculosis usually responds to medical therapy, surgical treatment is required when complications occur, including perforation, abscess, fistula, bleeding, complete intestinal obstruction or obstruction that fails to respond to conservative management.2 Caecal perforation due to tuberculosis is uncommon, and isolated caecal perforation with retroperitoneal involvement is particularly rare.3 Intestinal perforation may also occur at the beginning of, during or after anti-tubercular therapy, sometimes as part of a paradoxical inflammatory response.4-6 We report a rare case of retroperitoneal caecal perforation associated with ileocecal tuberculosis and multiple strictures, managed successfully by emergency surgery and continuation of anti-tubercular therapy.

Case presentation:

A 31-year-old woman presented to the emergency department in January 2025 with diffuse abdominal pain involving all quadrants, multiple episodes of vomiting, failure to pass flatus and stool, and low-grade fever for six days. She was a known case of abdominal tuberculosis and had been receiving anti-tubercular therapy for 10 days. There was no history of trauma or previous abdominal surgery.

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On examination, she had tachycardia with generalized abdominal tenderness and guarding. Routine blood investigations were within normal limits. Screening for HIV, HCV and HBsAg was negative. Erect abdominal radiography showed gas under the diaphragm. Ultrasonography of the abdomen showed dilated bowel loops with to-and-fro peristalsis, suggestive of intestinal obstruction.

Contrast-enhanced computed tomography of the abdomen demonstrated multifocal circumferential asymmetric wall thickening involving the terminal ileum, ileocecal valve, caecum and proximal ascending colon over a segment longer than 13 cm, with transmural perifocal infiltration, necrotic regional ileocolic lymphadenopathy, a small hyperdense perifocal collection, a skip lesion in the mid-ileal region and short-segment proximal obstruction. A tubercular etiology was considered most likely. Colonoscopy showed thickened ulcerated mucosa with luminal narrowing just beyond the hepatic flexure; the scope could not be negotiated beyond the narrowed segment. Biopsy from this area was suggestive of tubercular colitis.

In view of peritonitis with radiological evidence of perforation and obstruction, emergency exploratory laparotomy was performed. Intraoperatively, a 1 x 1 cm caecal perforation was identified, with multiple adhesions, most marked around the ileocecal junction. Two completely non-passable strictures were present approximately 30 cm and 70 cm proximal to the ileocecal junction. No abnormality was detected in the other abdominal organs. Right hemicolectomy with resection of diseased bowel, side-to-side jejuno-colic anastomosis and peritoneal toileting was performed. The specimen was sent for histopathological examination.

Figure 1. Postoperative specimen demonstrating caecal perforation.

Histopathology revealed non-caseating epithelioid granulomas surrounded by a rim of transformed lymphocytes, along with Langhans-type giant cells. The lamina propria showed dense lymphoid infiltration, consistent with granulomatous ileitis, most probably tuberculous ileitis. Chest radiography was normal, the Mantoux test was negative, erythrocyte sedimentation rate was 18 mm in the first hour and sputum CBNAAT was negative.

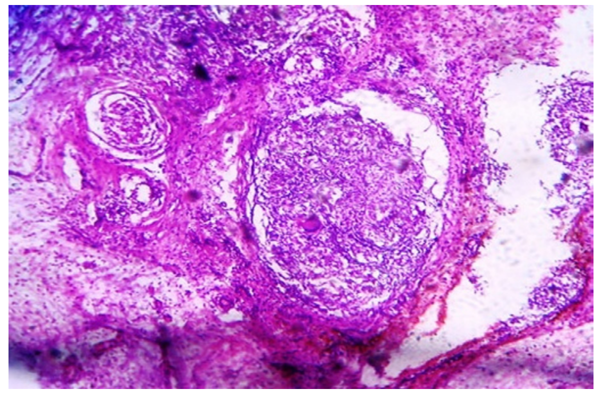

Figure 2. Microscopic appearance on histopathology showing granulomatous inflammation.

Anti-tubercular therapy was continued postoperatively with isoniazid, rifampicin, pyrazinamide and ethambutol for the intensive phase, followed by isoniazid and rifampicin for the continuation phase. The patient showed satisfactory postoperative recovery with overall clinical improvement and remained under fortnightly follow-up.

Discussion:

The present case is notable for the coexistence of caecal perforation, ileocecal adhesions and two tight proximal strictures in a patient recently started on anti-tubercular therapy. Caecal perforation is more commonly associated with conditions such as diverticular disease, inflammatory bowel disease, closed-loop obstruction, ischemia, trauma and malignancy; tuberculosis is an uncommon cause.3 In intestinal tuberculosis, strictures and obstruction are frequent surgical presentations, whereas perforation is less common but carries significant morbidity and mortality.

Ileocecal tuberculosis is a diagnostic challenge because symptoms are often nonspecific and microbiological tests may be negative. In this patient, chest radiograph, Mantoux test and sputum CBNAAT did not provide supportive evidence; however, the combination of CT findings, colonoscopic biopsy and postoperative histopathology supported the diagnosis. The presence of epithelioid granulomas with giant cells in the appropriate clinicoradiological setting is an important clue, although histology should always be interpreted along with clinical, radiological and microbiological findings.

Perforation during the early phase of treatment has been described in intestinal tuberculosis and may reflect progression of an undetected lesion, transmural necrosis, obstruction-related pressure effects or a paradoxical inflammatory response to anti-tubercular therapy.4-6 The risk of delayed diagnosis is high because worsening abdominal symptoms may initially be attributed to drug intolerance, nonspecific gastroenteritis or subacute obstruction.

Surgery in abdominal tuberculosis should be individualized. Current evidence supports operative intervention for acute complications such as free or contained perforation, abscess, fistula, massive bleeding, complete obstruction and obstruction that does not respond to medical treatment.2,7,8 Resection of the diseased segment with anastomosis is commonly performed when the patient is physiologically stable and local contamination is controlled; diversion may be required in unstable patients, gross contamination or poor bowel condition. Anti-tubercular therapy remains essential after surgery and should be continued according to national or international tuberculosis treatment guidance.9

This case emphasizes that a high index of suspicion must be maintained in endemic regions when a patient with suspected or confirmed abdominal tuberculosis develops acute abdomen, pneumoperitoneum or intestinal obstruction, even shortly after initiation of therapy. Early surgical exploration, adequate resection of diseased bowel, histopathological confirmation and continuation of anti-tubercular therapy can result in favorable outcome.

Conclusion:

Retroperitoneal caecal perforation is a rare and severe complication of abdominal tuberculosis. Because clinical manifestations may be nonspecific and routine tests can be negative, diagnosis requires integration of clinical suspicion, imaging, endoscopic biopsy and histopathology. In patients presenting with perforation and obstructing strictures, timely surgical intervention combined with appropriate anti-tubercular therapy is the key to successful management.

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